

AI Tutor for Personalized Student Learning Recommendations

Introduction:

Artificial Intelligence is widely used in education to provide personalized learning experiences. An AI Tutor analyzes student performance and recommends suitable learning materials based on strengths and weaknesses. It supports learning through quizzes, feedback, and progress tracking.

1. PEAS Framework

Performance Measure:

- Recommendation accuracy
- Student improvement
- Fast response time
- Student satisfaction

Environment:

Online learning platforms where students access educational content through computers or mobile devices.

Actuators:

- Display learning recommendations
- Generate personalized quizzes
- Send notifications and reminders
- Show progress reports

Sensors:

- Quiz and test scores
- Login and Activity records
- Time spent on learning material
- Student feedback and feedback

2. Environment Classification:

Partially Observable:

Student understanding cannot be observed directly.

Single-Agent:

The AI Tutor is the main decision-making agent.

Stochastic:

Learning outcomes are uncertain.

Sequential:

Current recommendations affect future performance.

Dynamic:

Student progress changes over time.

Continuous:

Learning develops gradually.

3.Critical Analysis**Most Difficult Dimension:**

Partially Observable environment is the biggest challenge because the AI Tutor must estimate student understanding from quiz results and activity data.

Utility Function:

$$U = 0.7 \times \text{Recommendation Accuracy} - 0.3 \times \text{Response Time}$$

This function gives higher importance to recommendation accuracy while also considering response time. Increasing the accuracy weight improves recommendation quality but may require more processing time.

4.JSON Structured Representation:

```
{
  "system_name": "AI Tutor for Personalized Learning Recommendations",
  "peas": {
    "performance_measures": [
      "recommendation_accuracy",
      "student_improvement",
      "fast_response_time",
      "student_satisfaction"
    ],
    "environment": "Online learning platform where students use mobile phones,
tablets, and computers to access educational content.",
    "actuators": [
      "display_recommendations",
      "generate_quizzes",
      "send_notifications",
      "show_progress_reports"
    ],
    "sensors": [
      "quiz_scores",
      "login_activity",
      "learning_time",
      "student_feedback"
    ]
  }
}
```

```

},
"environment_classification": {
  "observability": {
    "choice": "Partially Observable",
    "justification": "The tutor cannot directly observe student understanding and relies
on performance data."
  },
  "agents": {
    "choice": "Single-agent",
    "justification": "The AI Tutor is the main decision-making agent."
  },
  "determinism": {
    "choice": "Stochastic",
    "justification": "Student learning outcomes are uncertain and influenced by many
factors."
  },
  "episodic_sequential": {
    "choice": "Sequential",
    "justification": "Current recommendations affect future learning outcomes."
  },
  "static_dynamic": {
    "choice": "Dynamic",
    "justification": "Student knowledge and behavior change over time."
  },
  "discrete_continuous": {
    "choice": "Continuous",
    "justification": "Learning progress changes gradually over time."
  }
},
"utility_function": "U = 0.7*RecommendationAccuracy - 0.3*ResponseTime"
}

```